DiffCAM: Differentiable Core Status and a Research Plan for RSS

Repository audit and proposed research program

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Abstract

DiffCAM already contains a differentiable, history-dependent material-removal kernel, continuous delta actions, soft geometry objectives, hard-carve evaluation, and a G-code export/parse path. These support a focused robotics question: when do gradients through a relaxed irreversible geometric process improve an executable machining trajectory? The strongest prospective contribution is a controlled study, and ultimately an algorithm, for managing disagreement between smooth optimization and hard execution under structured CAM initialization. Neither arbitrary precision, autonomous safety, calibrated cutting physics, nor physical validation follows from the present implementation. This report audits those distinctions, preserves the useful findings of the autoresearch work, and specifies an experiment and release plan with explicit decision gates.

1 Scope and evidence provenance

The source snapshot is branch `autoresearch`, commit `41b7355f05a6315bf2af47ee478501e519ca9232`. Branch `core` was created from that snapshot. Evidence comes from source inspection, local experiment notes and logs, the supplied Method section, and the supplied *Differentiable CNC Simulator Paper Framing.md*. The framing prose is preserved in `docs/framing-source.txt`; its embedded formula images are not reproduced. It is a research proposal, not an independently verified survey. The cleanup inventory is `docs/cleanup-manifest.txt`. The complete local experiment directory was moved to `../backups/diffcam-core-2026-11-08/autoresearch/`; historical tracked files remain recoverable from the source commit. Local runs, credentials, and nested Overleaf repositories were not deleted.

The supplied MachineBench description concerns a separate evaluation system. Its archive counts, collision validation, leaderboard behavior, and Fusion probe are author-provided context, not results reproduced in this repository audit. This distinction is essential: exporting NC from DiffCAM does not establish that MachineBench's independent evaluator accepts or safely executes it.

2 What the implementation actually does

2.1 Representation, recurrence, and differentiation

`simulator/csg_simulator.py` implements `CSGSimulatorDelta` in Taichi. Tool and target primitives are analytic implicit functions, but stock is stored on a dense grid with a time axis. With

negative values denoting solid,

$$\phi_{t+1}(x_i) = \max\{\phi_t(x_i), -d_{C_t}(x_i)\}, \tag{1}$$

$$\widetilde{\phi}_{t+1}(x_i) = \mathrm{smax}_\kappa(\widetilde{\phi}_t(x_i), -\widetilde{d}_{C_t}(x_i)), \tag{2}$$

$$\mathrm{smax}_\kappa(a,b) = m + \kappa^{-1}\log(e^{\kappa(a-m)} + e^{\kappa(b-m)}), \quad m = \max(a,b). \tag{3}$$

Here C_t is the segment's approximated swept cylindrical cutter, and $\kappa = k/\texttt{k_ref}$. The stabilized log-sum-exp avoids exponential overflow. Spatial grid updates are parallel; the history and stock-dependent motion updates are sequential. Analytic primitives do not make the state mesh-free or resolution-independent. Moreover, CSG combinations and several primitive functions are not exact Euclidean signed distances everywhere.

Stock values and adjoints alone require approximately $8(T+1)N_xN_yN_z$ bytes in float32, before other fields and runtime allocations. Resolution refinement and long horizons therefore remain coupled memory costs. The current stock-only allocation is a useful engineering choice: the larger machine envelope is metadata, not a voxelized workspace.

Delta parameters reconstruct positions through $p_{t+1} = p_t + \alpha_t\Delta_t$. The scale α_t clips requested speed against feed/rapid limits in physical units. The regime is selected by a hard stock-clearance test at the commanded destination. This is a piecewise geometric execution model, not the machine's full servo dynamics. The CAM planner separately models acceleration-limited exact-stop motion; training and posted execution are therefore not identical dynamics. The claim that absolute coordinates cause severe instability still needs a matched ablation.

2.2 Loss and the safety mismatch

The implemented terminal loss averages pointwise squared soft occupancy errors:

$$o_i = \sigma(-\widetilde{\phi}_T(x_i)), \quad q_i = \sigma(-d_{\mathrm{target}}(x_i)), \tag{4}$$

$$L_{\mathrm{geom}} = \frac{1}{N}\sum_i \left[w_g\{q_i(1-o_i)\}^2 + w_r\{(1-q_i)o_i\}^2\right]. \tag{5}$$

This differs from squaring aggregate gouge and residual volumes. Occupancy uses an approximately one-voxel sigmoid scale; k controls smooth geometry, rather than directly annealing this occupancy sigmoid. Optional annular weighting and loss shifts exist. Path proximity, length, jerk, step, air, and tool-gouge penalties are available, with many disabled by default.

The holder objective is not the single-point expression in the supplied Method:

$$L_h = \frac{w_h}{N}\sum_{t,i}\sigma(-\widetilde{\phi}_{t+1}(x_i))[\max(0, m_h - d_{H_t}(x_i))]^2. \tag{6}$$

It integrates squared penetration of the swept holder over soft remaining stock, using *post-cut* stock. This can credit material removed during the same segment when checking its holder. It does not certify chronological clearance of the full assembly. A finite soft penalty cannot guarantee safety; rapid clearance, fixture collision, shank geometry, and controller semantics require separate validation. Use "collision-penalized" until independent hard checks establish the relevant property.

2.3 Swept-volume and export limitations

Both hard and soft cutter functions choose the segment parameter by closest projection in XY, then interpolate tool height at that parameter. Pure vertical motion has a zero XY direction

and effectively selects the initial endpoint; it does not cover the complete vertical sweep. For simultaneous XYZ motion, closest XY is not generally the minimizer over the full cylindrical sweep. Consequently hard Boolean subtraction is exact *for the supplied implicit operand*, not necessarily for the physical swept cutter. Existing comments claiming universal subdivision invariance should be read as an intended property; horizontal fixed-Z tests alone cannot establish it for general paths. This is a publication-blocking geometry issue, especially for plunges and holes.

`cam/` contains physical units, stock/work-origin transforms, Haas and RS274 posts, a modal motion parser, trajectory planning, path metrics, and hard carve execution. The Haas post adds approach/retract motion. A round trip through the local parser is valuable regression evidence but shares assumptions with the exporter. It is not independent controller validation or a machining trial. All added approach, retract, rapid, and unit conversions must be graded from the actual posted file.

2.4 Capability and claim ledger

Capability	Evidence now	Required before publication claim
Differentiable removal	Taichi tape through soft cuts and deltas	Gradient checks across shapes, motion regimes, horizons and scales
Hard checkpoint selection	Present historically; retained in minimal runner	Independent NC score, fixed validation rule, no test-set selection
CAM initialization	Historical contour, cavity, shell and adaptive heuristics	Reusable seed generation, unmodified-seed baseline and held-out geometries
Safety	Soft holder penalty and geometric diagnostics	Pre-removal chronological swept tests, adversarial fixtures and fixture model
Process quality	Air/time and heuristic force/breakage fields	Calibrated force/engagement model and controller timing validation
Physical execution	Export/parse/planner support	Measured parts, real cycle time and matched baseline trials
MachineBench integration	Described in supplied manuscript	Versioned independent evaluator and retained NC records

3 What autoresearch contributes

The archived `idea.md` reports a dual-adaptive contour scheme: angular and vertical cross-section variation select smoothing and finishing behavior without directly branching on the shape name. Its historical best hard-Dice values are reproduced below as *unreproduced log claims*, not means, confidence intervals, or comparative publication results.

Target	Reported best hard Dice	Interpretation
Cylinder	0.8898	Contour plus finishing
Sphere	0.8311	Finishing disabled
Box	0.7596	Softer initialization
Pyramid	0.7976	Contour initialization
Sphere bowl	0.6471	Cavity initialization
Sphere hole	0.2630	Persistent failure; later variants also logged

The note attributes these to experiments near commits `02fb503` and `845999e`; provenance and exact configurations must be reconstructed before reuse. It also reports benefits from hard selection and annealing, and geometry-dependent regressions from finishing passes. These motivate hypotheses, not numerical claims of superiority. The TSV's column named `dice` is not consistently a hard score: one pyramid entry gives 0.873104 while its description explicitly says hard Dice 0.277. Do not pool that column.

Promising existing ideas are (i) target-derived topology priors, (ii) morphology- conditioned smoothing/finishing, (iii) history-dependent stock and holder coupling, (iv) saved intermediate stock for staged optimization, and (v) deployment-aware checkpoint selection. Preference feedback and breakage proxies are exploratory extensions; they should not expand the central paper before geometric validity is established. A sphere with inaccessible undercuts is also not automatically a feasible one-setup three-axis target. Separate accessibility limits from optimizer failures with reachable target definitions and an oracle feasibility check.

4 Prior work and defensible positioning

The framing memo's broad "first differentiable CNC" claim is not defensible. CNC-Net already learns CNC operation parameters from target shapes [1]; CNCGEN explicitly couples operation planning, intermediate stock, and learned verifier feedback [2]. DiffCAM's candidate distinction is explicit analytic removal derivatives for refining a fixed target's executable trajectory, without a learned removal verifier. Demonstrating that distinction requires implementation and equal-budget experiments, not terminology alone.

DiSECt differentiates FEM/contact/damage for soft-material cutting and optimizes force-related controls [3]. It provides a robotics precedent, but DiffCAM currently models rigid removal geometry rather than fracture or calibrated cutting dynamics. PlasticineLab motivates comparing gradients and RL on long-horizon shape manipulation [4]. Neither establishes that one optimizer must win on CNC machining.

Suh et al. analyze why stiffness and discontinuities can undermine first-order estimators [5]. Pang et al. connect analytic and randomized smoothing in contact-rich planning [6]. These motivate studying the analogous tradeoff in irreversible stock removal; the equivalence does not automatically transfer to DiffCAM's surrogate or to PPO training.

Implicit neural-field process planning [7], four-axis continuous path optimization [8], and design-for-machinability optimization [9] delimit the landscape: implicit fields, continuous path refinement, and manufacturing-aware gradients are already established. The memo's Lichter and Hadhazi item could not be independently verified in this audit's primary- source search; do not use its numerical comparisons or claimed novelty boundary until its full text and publication metadata are obtained. TopoCut and the remaining specialized CAM/RL references likewise need individual full-text verification before their detailed claims enter a submission.

Recommended thesis. *History-aware differentiable machining can efficiently refine structured CAM trajectories, provided that smoothing bias and execution disagreement are measured and controlled against hard posted-path evaluation.* This is a testable hypothesis, not a conclusion. Suggested title: *DiffCAM: Controlling the Relaxation-to-Execution Gap in Differentiable Machining.*

Candidate algorithmic contribution. Develop a hard-validated continuation method: optimize within a trust region around a CAM seed, measure soft/hard/posted disagreement, and adjust smoothing, step size, or topology when disagreement or stagnation exceeds a validation-set threshold. Hard selection alone is a useful baseline, not sufficient novelty. Compare this proposed method against fixed annealing with identical evaluation budgets. Preserve and ablate the existing morphology-based schedule as a distinct alternative. Topology search followed by gradient refinement is an optional extension; do not claim a hybrid method before implementing it.

5 A falsifiable experimental program

5.1 Stage 0: establish a trustworthy oracle

Before optimizer sweeps, fix and test pure-Z and sloped swept cutters against independent analytic fixtures. Include horizontal slots of known volume, vertical plunges, diagonal entries, repeated cuts, reversals, tool-length changes, narrow pockets, and legitimate through-holes. Test subdivision and permutation invariance of final hard geometry where applicable; safety remains order-sensitive. Cross-check hard carving against an independently implemented NC evaluator and refine its resolution until numerical uncertainty is below the claimed effect. Test *pre-cut* holder occupancy and rapid interiors, including endpoints that are clear while the segment crosses solid stock. Correct the holder's chronology and validate shank/holder geometry before asserting collision freedom.

Check autodiff against central differences of the *same smooth objective* away from gates, and report behavior at gates separately. Sweep finite-difference step sizes to distinguish truncation from float32 cancellation. A gradient of a hard voxel score is often zero almost everywhere; estimate gradients of an explicitly randomized hard objective when studying bias/variance. Compare the same perturbation distribution and objective, and report estimator uncertainty.

5.2 Stage 1: initialization and solver study

Build three task families: stepped/asymmetric facing; pockets with islands and narrow channels; and variable-width slots/deep pockets. Use multiple dimensions and held-out topologies per family. Current analytic primitives are diagnostic fixtures, not sufficient coverage. Supply facing raster, contour offset, and adaptive/trochoidal seeds, with feasibility checked under the same cutter, stock, setup, motion budget and tolerances. Target-generated seeds already solve part of the geometry problem, so their unoptimized score must be reported.

The minimum comparison is nominal CAM; CAM plus hard CEM; random/line plus smooth Adam; CAM plus smooth Adam with fixed k; and CAM plus annealed Adam with hard selection. Add morphology-adaptive and disagreement-adaptive variants only after those baselines work. Residual PPO on the hard simulator directly tests the framing memo's RL question, but requires a restored, controlled experiment layer outside the minimal runtime dependencies. If retained, compare both end-to-end training cost and amortized per-part inference; separate policy learning across parts from per-instance trajectory optimization.

Use paired seeds and identical initial paths, trajectory degrees of freedom, constraints and task splits. Five seeds per cell is a starting minimum, not a power guarantee. Record simulator calls,

gradient calls, measured wall time, peak memory, hardware, warm-up and compilation separately. Give both equal-time and equal-simulation-budget results. The initial pilot determines attainable sample size and practical effect thresholds before freezing the test protocol.

5.3 Stage 2: mechanism and ablations

Sweep k, annealing schedule, temporal segmentation, grid spacing, and horizon. Measure hard IoU, soft/hard Dice disagreement, gouge, residual, gradient norm, finite-difference agreement, and optimization progress. Repeat one geometric path at several segment counts to isolate representation bias from better path coverage. For exact log-sum-exp aggregation over M fixed operands,

$$0 \leq \operatorname{smax}_\kappa(a_1, \ldots, a_M) - \max_i a_i \leq \frac{\log M}{\kappa}. \tag{7}$$

This motivates a segmentation/smoothing study. It is *not* a complete error bound for this code: tool operands are also smoothed and stock-dependent speed clipping changes the path. Distinguish operator bias at a fixed path from trajectory drift under coupled dynamics.

Ablate delta versus absolute positions with matched executed paths, constraints and parameter scales; hard versus soft selection; geometry-only versus holder penalties; fixed versus morphology-conditioned schedules; and full versus truncated adjoints with saved stock. No universal U-shaped curve or orders-of- magnitude solver advantage should be presupposed. Report counterexamples, including narrow features, topology changes and inaccessible targets.

5.4 Stage 3: posted NC and physical evidence

Grade the emitted file, including approach/retract moves, through MachineBench's independent NC-only path. The supplied description supports geometric fidelity, stock containment, holder clearance and rapid-stock checks for all sources; Fusion-specific engagement and sequencing remain unavailable for NC-only runs. Missing measurements must remain null, never a pass. The reported 51 real programs and thirteen adversarial cases belong to MachineBench validation and must be cited to its records; four flags associated with reduced-shank geometry need careful interpretation as a model limitation, not unqualified physical collision ground truth. The corrected contradictory synthetic fixture should retain its original record and corrected geometry.

Use wax trials to validate geometry and work offsets, then matched Al 6061-T6 trials under a fixed qualified machining setup. Measure actual cycle time and part deviations with CMM or scanning. Treat spindle load as a proxy unless calibrated against force instrumentation. No force reduction claim is supported by the current heuristic force/breakage fields. Compare nominal CAM and the chosen optimizer on identical tools, stock and material settings; do not run known-unsafe baseline trajectories merely to complete a table. Retain NC, controller settings, stock/fixture dimensions, tool geometry and measurement uncertainty with each physical record.

6 Evaluation and reporting contract

The primary outcome should be hard posted-path geometric quality subject to predeclared gouge and safety criteria. Report IoU, Dice, gouge/residual volumes and explicitly named denominators, maximum/p99 surface deviation in mm, clearance failure counts, cycle time and air-cut fraction. Current core metrics include Dice, ASD and HD95; those are not interchangeable with MachineBench's maximum/p99 metrics. For identical binary masks, $\text{IoU} = \text{Dice}/(2 - \text{Dice})$, but conversion does not resolve different grids, trajectories or definitions.

Use paired per-instance comparisons and task-stratified reporting; do not pool uneven task coverage. Continuous means may use t intervals when justified by the data, while binary rates use Clopper–Pearson intervals. At five successes in five trials, the two-sided 95% lower bound is approximately $(0.025)^{1/5} = 0.478$: a perfect observed rate is weak evidence of universal safety. Show all trial records, seeds, exclusions and failures. Exclude infrastructure/document-compromised trials under a predeclared rule, while retaining valid algorithm failures in the outcome analysis.

MachineBench's win ranking and IoU ≥ 0.5 efficiency gate can be retained for compatibility, but the RSS paper should prioritize effect sizes and uncertainty over leaderboard wins. Its Fusion collision flag means "no collision reported" until fault-injection validation. Holder, rapid and containment checks are separate outcomes, including when not ranked. Material-blind geometric scores do not establish shop readiness. Do not transplant LLM token/cache metrics into a solver comparison without a scientific reason.

7 Milestones, stop conditions, and manuscript plan

Window	Deliverable	Exit gate
Weeks 1–2	Sweep geometry corrections, gradient tests, chronological safety fixtures, evaluator contract	Known-volume and motion fixtures pass; discrepancy quantified
Weeks 3–4	Feasible task suite, reusable CAM seeds, nominal and hard-CEM baselines, pilot	Frozen configurations, splits, budgets and effect thresholds
Weeks 5–6	Paired solver comparison and smoothing/segmentation/horizon ablations	Reproducible effects on held-out parts; failures disclosed
Weeks 7–8	Hard-validated continuation and independent NC evaluation	Improvement beyond ordinary annealing/selection at matched budget
Weeks 9–10	Wax/Al trials, measured timing and geometry	Physical tolerances and predicted effects verified
Weeks 11–12	Artifact audit, figures, literature update, manuscript	Claims traceable to records; independent reproduction

These are effort estimates, not a promise to meet an unverified conference deadline. The retrieved official RSS 2026 call specifies eight main-text pages, excluding references [10]; the memo's five-to-six-page assumption is not a current submission rule. RSS 2026 is already past at this audit date. Verify the target year's official call, template, anonymity and artifact rules before committing to its deadline; this audit did not verify RSS 2027 requirements.

Allocate roughly one page to the problem and contribution, one to related work and formulation, two to the mechanism/algorithm, three to evidence, and one to limitations and discussion, adapting to the actual call. Key figures are: (1) hard/soft/posted stock disagreement along one trajectory; (2) quality versus compute with unoptimized CAM anchors; (3) segmentation–sharpness and horizon ablations; (4) physical part measurements with matched NC and timing.

If geometric corrections erase the historical gains, report that finding and rebuild the experiment rather than preserving the old headline. If continuation adds nothing beyond fixed annealing, frame the contribution as a validated simulator and mechanistic study only if its insight is strong enough. If physical or force validation is unavailable, narrow the claims explicitly; a benchmark integration alone does not establish a robotics algorithm contribution.

8 Core release and reproducibility

The reduced tree retains the Taichi kernel, geometry metrics, CAM bridge, CAM regression fixture, and a small NumPy Adam runner. RL environments/trainers, preference dashboards, cluster/bootstrap tooling, obsolete diagnostics and GUI rendering were removed. Optional differentiable penalties and saved-state APIs remain because they support the proposed mechanism experiments. The historical large training harness and seed heuristics remain on `autoresearch`; the new runner accepts externally supplied seed trajectories and is not a drop-in reproduction of historical experiments.

`algorithms/optimize.py` records configuration, soft/hard score histories, selected hard-score checkpoint, deltas, executed hard-path positions and both NC dialects. The smoke initializer is deliberately just a wiring test; its score is not a machining result. Preserve raw artifacts and hashes for future runs, including the initial trajectory, software/environment versions and evaluator revision. The cleanup validation passed 20 CPU tests, including a finite-difference gradient check and CAM carve round trip. A two-update optimizer smoke run exported and parsed both NC dialects. These are regression results, not research performance evidence. Normal dependency synchronization was blocked by DNS, so tests used compatible cached packages in an isolated environment. PDF compilation was unavailable because `pdflatex` is not installed. Full commands and limitations are recorded in `report/validation.txt`; the adjacent Makefile provides the report build.

References

[1] M. Yavartanoo, S. Hong, R. Neshatavar, and K. M. Lee. CNC-Net: Self-Supervised Learning for CNC Machining Operations. CVPR, 2024. https://arxiv.org/abs/2312.09925.

[2] X. Zhou, B. Lin, Y. Feng, and J. Zheng. CNCGEN: A Dataset and Framework for Machining Process Planning and Toolpath Generation from B-rep Models. Preprint, 2026. https://arxiv.org/abs/2609.39738.

[3] E. Heiden et al. DiSECt: A Differentiable Simulation Engine for Autonomous Robotic Cutting. RSS, 2021. https://arxiv.org/abs/2105.12244.

[4] Z. Huang et al. PlasticineLab: A Soft-Body Manipulation Benchmark with Differentiable Physics. ICLR, 2021. https://arxiv.org/abs/2104.03311.

[5] H. J. T. Suh, M. Simchowitz, K. Zhang, and R. Tedrake. Do Differentiable Simulators Give Better Policy Gradients? ICML, 2022. https://arxiv.org/abs/2202.00817.

[6] T. Pang, H. J. T. Suh, L. Yang, and R. Tedrake. Global Planning for Contact-Rich Manipulation via Local Smoothing of Quasi-dynamic Contact Models. https://arxiv.org/abs/2206.10787.

[7] Implicit Neural Field-Based Process Planning for Multi-Axis Manufacturing: Direct Control over Collision Avoidance and Toolpath Geometry. Preprint, 2025, revised 2026. https://arxiv.org/abs/2511.17578.

[8] Z. Zhang et al. Continuous Toolpath Optimization for Simultaneous Four-Axis Subtractive Manufacturing. Computer Graphics Forum, 2024. https://haisenzhao.github.io/FourAxis/index.html.

[9] H. Chen and L. B. Kara. Design for Machinability: Topology Optimization for Subtractive Machining Using Trimmed Neural Fields. Preprint, 2026. https://arxiv.org/abs/2610.15301.

[10] Robotics: Science and Systems. Official 2026 Call for Papers. Accessed 8 November 2026. https://roboticsconference.org/information/cfp/index.html.